DELIVERABLE 4

Presentation Content

CareFlow — Clinical Pathway Process Mining

13 slides · copy-paste ready for PowerPoint
Presenter: Yash Rahangdale · **Ref. No.:** ITS/DA/1393
Date: 10 October 2026

Kaise use karein: PowerPoint kholiye → New Presentation → har slide ke liye **Insert → New Slide (Title and Content)** → neeche diya Title aur Bullets copy-paste kar dijiye. Charts ke liye `outputs/figures/` folder se PNG drag-drop kar dijiye (slide ke saath figure ka naam likha hai). Total time: 10–12 minute.

SLIDE 1 — Title slide

Title	CareFlow: Clinical Pathway Process Mining
Subtitle	Discovering, Analysing and Predicting Hospital Patient Pathways from Event Logs
Presenter line	Yash Rahangdale · Data Associate L1 Intern · Data Analytics Domain
Footer line	Infotact Solutions · Ref. No. ITS/DA/1393 · 10 October 2026
Figure	koi nahi (clean title slide)

SLIDE 2 — Agenda

Title	Agenda
Bullets	1. Problem: why clinical pathways drift 2. What is process mining and why healthcare needs it 3. Data: 1,500 patient journeys, 13,583 events 4. Discovered process model and pathway variants 5. Performance analysis: bottlenecks, cost, resources, load

	6. Conformance checking 7. Predictive model: who will become a long stay? 8. Dashboard demonstration 9. Business recommendations, limitations and next steps

SLIDE 3 — Problem statement

Title	The Problem: pathways are designed one way, executed another
Bullets	• A patient journey runs Registration → Triage → Tests → Diagnosis → Treatment → Discharge • In reality every journey differs: repeated tests, ICU escalation, delayed discharge • Traditional dashboards show aggregates — bed occupancy, average stay, monthly cost • Aggregates reveal **that** something is slow, never **where inside the process** • Result: waiting time and cost accumulate invisibly, and nobody owns the bottleneck

Bolne ke liye: “Ek line me problem — hospital ko pata hota hai ki patient zyada der ruk raha hai, lekin yeh nahi pata ki process ke andar exactly kahan ruk raha hai.”

SLIDE 4 — What is process mining

Title	Solution Approach: Process Mining on the Event Log
Bullets	• An **event log** records every activity: case_id + activity + timestamp • **Discovery** — derive the real process model from data, no interviews needed • **Conformance** — measure how far reality deviates from the expected pathway • **Enhancement / prediction** — quantify bottlenecks and forecast outcomes • Implemented in Python with pandas, networkx, scikit-learn, Streamlit; pm4py adds the official miner algorithms

SLIDE 5 — Data

Title	Dataset: 6 months of hospital activity
Bullets	• 1,500 patient cases · 13,664 raw events → 13,583 after cleaning • 15 clinical activities · 28 resources · 10 departments • Window: 01 Mar 2026 → 31 Aug 2026 • Required columns: case_id, activity, timestamp (+ optional resource, cost, durations) • Cleaning: 81 duplicate events removed, 203 blank resource

	fields filled • Four clinical pathway templates: Outpatient 35 %, Emergency Surgical 25 %, Chronic Follow-up 25 %, ICU Escalation 15 %
Figure	`case_volume_trend.png`

Honesty point (ise zaroor bolein): “Dataset synthetic hai — assignment ki confidentiality constraint aur 1-month window ke kaaran. Generator realistic distributions, rework loops aur data-quality defects inject karta hai, aur poori pipeline seed = 42 par reproducible hai.”

SLIDE 6 — Methodology / pipeline

Title	Methodology: a seven-stage pipeline
Bullets	1. Event-log generation / import 2. Cleaning + feature engineering (9 engineered features) 3. Process discovery — Directly-Follows Graph + variant ranking 4. Performance analytics — LOS, bottleneck, cost, resources, temporal load 5. Conformance checking — token-based replay 6. Predictive modelling — three algorithms, two feature sets 7. Interactive Streamlit dashboard — 5 tabs, 4 live filters • Runtime: ~17 seconds end-to-end · fully reproducible

SLIDE 7 — Discovered process model

Title	Discovered Care Pathway
Bullets	• 28 directly-follows edges across 15 activities • All 1,500 cases start at Registration and end at Discharge — a well-formed log • 14 distinct pathway variants • Top variant = 29.87 % of cases · Top 5 variants = 80.94 % • Interpretation: variation is **concentrated** — standardising five variants covers most of the process
Figure	`dfg_careflow.png` (full slide bhar ke dikhaiye)

SLIDE 8 — Variants

Title	Pathway Variants: where patients actually go
Bullets	• 14 unique activity sequences across 1,500 journeys • Top 5 variants explain 80.94 % of all cases • Remaining 9 variants are exceptional journeys (ICU escalation, complex surgical cases) • Every additional variant means extra coordination, extra training and extra cost

Figure	`variants_top10.png`

SLIDE 9 — Performance analytics

Title	Performance Analysis: the numbers that matter
Bullets	• Average length of stay **9.65 h** (median 6.14 h, P90 18.69 h) — right-skewed • Long-stay cases (> 24 h): **17 (1.13 %)** • Biggest bottleneck: **Observation — 226.61 min average wait** • Total waiting time across all cases: **868,869.6 minutes** of lost capacity • Total treatment cost **INR 2,58,86,760** · average **INR 17,258** per case • Costliest department: **Operation Theatre** • Busiest resource: **REC-01** with 1,038 events • Peak load: **Friday ≈ 14:00**
Figures	`bottleneck_waiting.png` + `kpi_los_distribution.png`

SLIDE 10 — Pathway comparison

Title	Cost follows the pathway, not the volume
Bullets	• P1 Outpatient — 521 cases · 4.52 h · INR 5,150 • P3 Chronic Follow-up — 400 cases · 5.77 h · INR 7,170 • P4 ICU Escalation — 204 cases · 15.15 h · INR 22,282 • P2 Emergency Surgical — 375 cases · 17.94 h · INR 42,107 • Emergency Surgical costs **8.2×** Outpatient per patient on fewer patients • Implication: staffing and budgeting must be pathway-aware
Figure	`pathway_comparison.png`

SLIDE 11 — Conformance and prediction

Title	Conformance Checking and Long-Stay Prediction
Bullets	**Conformance (token-based replay):** • Model = 28 DFG edges (occurrence ≥ 3) • Average fitness 1.0000 · 100 % cases conforming • Explanation: synthetic data is generated from fixed templates, so the model can replay its own data perfectly; on a real hospital log expect **0.70–0.95** **Prediction:** • Target: length of stay > 24 hours (1.13 % positive rate) • Three algorithms × two feature sets (full 19 vs leakage-free 14) • Production model: **Gradient Boosting** — accuracy 0.997 ·

	precision 1.000 · recall 0.667 · F1 0.800 · ROC-AUC 0.998 • Zero false alarms on the test set • Top drivers: number of events, unique activities, ICU transfer, rework count
Figures	`conformance_fitness.png` + `model_roc.png`

SLIDE 12 — Dashboard demonstration

Title	Interactive Dashboard (live demonstration)
Bullets	• Built with Streamlit + Plotly — runs with `streamlit run dashboard/app.py` • 4 live sidebar filters: care pathway, department, length of stay, triage level • 5 KPI cards: cases, events, average LOS, average cost per case, average fitness • 5 tabs: Overview · Process Discovery · Conformance · ML Model · Data • 12+ interactive charts including LOS histogram, bottleneck ranking, cost treemap, DFG, ROC curve, confusion matrix, feature importance • Filtered data downloadable as CSV directly from the dashboard
Figures	Dashboard ke apne screenshots (Overview, Discovery, Conformance, ML tabs)

SLIDE 13 — Recommendations, limitations and conclusion

Title	Recommendations, Limitations and Next Steps
Bullets	**Recommendations** • Decongest Observation — second ward-round slot plus a dedicated discharge desk • Fast-track morning laboratory sampling (second-largest wait category) • Roster to the load curve: Friday ~14:00, late-morning to mid-afternoon band • Deploy the leakage-free long-stay flag at admission (zero false positives) • Standardise the top 5 variants, then re-run conformance quarterly **Limitations** • Synthetic reference data; conformance 1.0 is a property of the generator • Only 3 positive test cases — wide confidence interval on F1 and recall • DFG replay is less precise than Petri-net alignment **Next steps** • Integrate a real HIS/EMR feed (HL7 / FHIR), add class

	balancing and SHAP explanation, build a what-if simulator, add drift monitoring **Conclusion:** from raw event log to actionable, deployable insight — in one reproducible pipeline.

Speaker notes — 12 minute timing plan

Slides	Minutes	Focus
1–2	0:00 – 1:00	Title, agenda — brisk
3–4	1:00 – 2:30	Problem and approach
5–6	2:30 – 4:00	Data and methodology
7–8	4:00 – 6:00	Process model and variants — the core contribution
9–10	6:00 – 8:00	Analytics and business numbers
11	8:00 – 10:00	Conformance and prediction
12	10:00 – 11:30	Live dashboard demonstration
13	11:30 – 12:00	Recommendations and close

Anticipated questions

Question	Answer
Data asli hai?	Reference run synthetic hai, confidentiality constraint ke kaaran. Pipeline real log par `python run_all.py --skip-data` se chal jaata hai.
Fitness 100 % kyun?	Synthetic generator ke saare paths model me cover hote hain. Real data par 0.70–0.95 expected hai.
Leakage-free kya hai?	Outcome ke baad bante duration columns model se hata diye, warna 1.000 accuracy fake hoti.
Business impact?	Bottleneck identified (Observation, 226.6 min), roster peaking Friday 14:00, long-stay flag at admission.
Scale ho sakta hai?	13,583 events 17 seconds me process hote hain; pandas-based, million-row logs tak vertically scalable.

End of presentation content — 13 slides, 12-minute delivery. Yash Rahangdale · Ref. No. ITS/DA/1393 · Infotact Solutions · 10 October 2026.